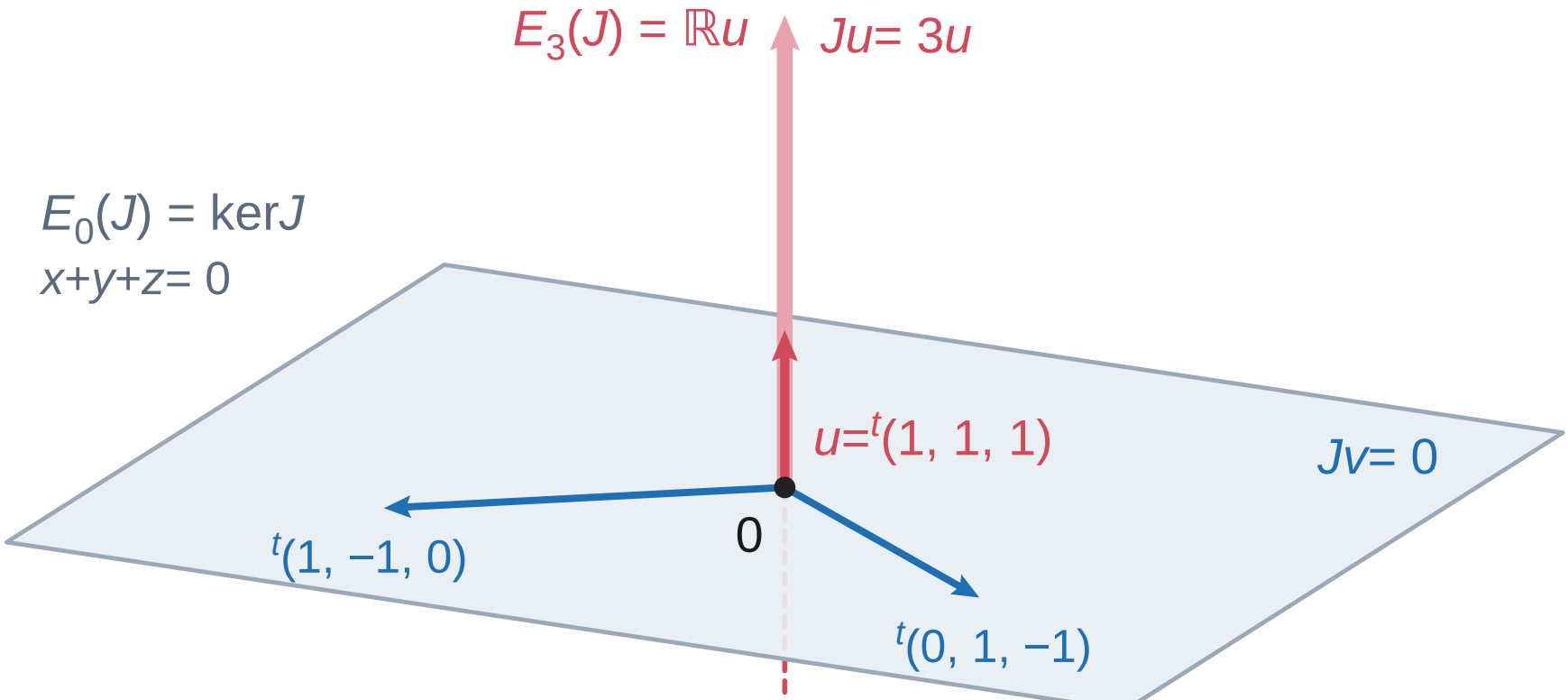
$E_3(J) = \mathbb{R}u$
$Ju = 3u$
$E_0(J) = \ker J$
$x+y+z = 0$
$u = {}^t(1, 1, 1)$
$Jv = 0$
0
${}^t(1, -1, 0)$
${}^t(0, 1, -1)$